Multi-cluster flocking in the Motsch-Tadmor model

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Collective motions

Flocking means the velocity of individuals in a group converges to a common on by using limited information and simple rules, such as the flocking of birds and swarming of fish.

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Flocking

Characteristics:

- the particles are in a bounded domain,
- the velocity difference of particles is zero.

Mathematical description: Denote by $\mathbf{x}_i$, $\mathbf{v}_i$ the position and velocity of the i-th particle in phase space $\mathbb{R}^d_x \times \mathbb{R}^d_v$.

Definition: Flocking

We call an ensemble $\mathcal{G} := \{(\mathbf{x}_i(t), \mathbf{v}_i(t))\}_{i=1}^N$ tends to flocking asymptotically if

$$\sup_{t \geq 0} |\mathbf{x}_i(t) - \mathbf{x}_j(t)| < \infty \quad \text{and} \quad \lim_{t \to \infty} |\mathbf{v}_i(t) - \mathbf{v}_j(t)| = 0.$$

Natural questions

- *Models to describe flocking*
- *Conditions for flocking, including sufficient and necessary conditions*

Necessary conditions for flocking: conditions for non-flocking

Definition: Non-flocking

We can find $i_0 \neq j_0$ such that

$$\sup_{t \geq 0} |\mathbf{x}_{i_0}(t) - \mathbf{x}_{j_0}(t)| = \infty \quad \text{or} \quad \limsup_{t \to \infty} |\mathbf{v}_{i_0}(t) - \mathbf{v}_{j_0}(t)| > 0.$$

F. Cucker and S. Smale [Cucker, Smale, IEEE, 2007] proposed a model to describe the phenomenon of flocking:

$$\begin{cases} \dot{\mathbf{x}}_i(t) = \mathbf{v}_i(t), \quad t > 0, \quad i = 1, 2, \cdots, N, \\ \dot{\mathbf{v}}_i(t) = \displaystyle\sum_{j=1}^{N} a(|\mathbf{x}_j(t) - \mathbf{x}_i(t)|)\big(\mathbf{v}_j(t) - \mathbf{v}_i(t)\big), \\ (\mathbf{x}_i, \mathbf{v}_i)(0) = (\mathbf{x}_i^0, \mathbf{v}_i^0), \end{cases}$$

where

- $\mathbf{x}_i$, $\mathbf{v}_i$: the position and velocity of the i-th particle in phase space $\mathbb{R}^d_x \times \mathbb{R}^d_v$,
- $a(s) = \frac{H}{(1+s^2)^\beta}$: the communication weight with fixed constant $H > 0$ and $\beta \geq 0$,
- $|\mathbf{x}| := \sqrt{\sum_{j=1}^d |x^j|^2}$.

Sufficient conditions for flocking

F. Cucker and S. Smale showed:

(i) If $\beta < \frac{1}{2}$, flocking emerge for any given initial data, i.e.,
$\Longrightarrow$ Unconditional flocking.

(ii) If $\beta \geq \frac{1}{2}$ and the initial satisfies a given explicit relation, flocking emerge, i.e.,
$\Longrightarrow$ Conditional flocking.

S.-Y. Ha and J. Liu [Ha, Liu, CMS, 2009] considered the Cucker-Smale model with general communication weight:

$$\text{(C-S Model)} \quad \begin{cases} \dot{\mathbf{x}}_i(t) = \mathbf{v}_i(t), \quad t > 0, \quad i = 1, 2, \cdots, N, \\ \dot{\mathbf{v}}_i(t) = \dfrac{\kappa}{N} \displaystyle\sum_{k=1}^{N} \psi(|\mathbf{x}_k(t) - \mathbf{x}_i(t)|)\big(\mathbf{v}_k(t) - \mathbf{v}_i(t)\big), \\ (\mathbf{x}_i, \mathbf{v}_i)(0) = (\mathbf{x}_i^0, \mathbf{v}_i^0), \end{cases}$$

where

- κ: the positive coupling strength,
- ψ: the communication weight satisfying

$$\psi(s) > 0, \quad (\psi(s_2) - \psi(s_1))(s_2 - s_1) \leq 0, \quad \text{for all } s, s_1, s_2 \geq 0.$$

They showed: if

$$\kappa > \frac{|\mathbf{v}^0 - \mathbf{v}_c^0|}{\int_{|\mathbf{x}^0 - \mathbf{x}_c^0|}^{\infty} \psi(r)dr} =: \kappa_{upp}(\mathbf{x}^0, \mathbf{v}^0),$$

the flocking emerge asymptotically. More precisely, there exists a positive constant x_M such that

$$\sup_{t\geq 0} |\mathbf{x}(t) - \mathbf{x}_c(t)| \leq x_M, \quad |\mathbf{v}(t) - \mathbf{v}_c(t)| \leq |\mathbf{v}^0 - \mathbf{v}_c^0| e^{-\psi(2x_M)t},$$

where $\mathbf{x}_c$ and $\mathbf{v}_c$ are the average of position and velocity respectively.

Conclusion:

- For a long-range communication weight (with infinite L^1-norm):

$$\kappa > 0 \quad \Longrightarrow \quad \text{Unconditional flocking.}$$

- For a short-range communication weight (with finite L^1-norm):

$$\kappa > \kappa_{upp}(\mathbf{x}^0, \mathbf{v}^0) \quad \Longrightarrow \quad \text{Conditional flocking.}$$

D. Ko, S.-Y. Ha, and Y. Zhang [Ha-Ko-Z, MMMAS, 2017] considered the Cucker-Smale model (C-S Model) with short-range communication weight, and showed, for the initial data

$$\max_{i \neq j} |\mathbf{v}_i^0 - \mathbf{v}_j^0| > 0,$$

there exists $\kappa_{low} = \kappa_{low}(\mathbf{x}_0, \mathbf{v}_0)$ such that if $\kappa < \kappa_{low}$, flocking does not occur.

Remarks:

- κ_{low} has a precise formula, and $\kappa_{low} < \kappa_{upp}$.
- The necessary condition for flocking is $\kappa > \kappa_{low}$.

Conclusion: For a short-range communication weight and any given initial configuration $(\mathbf{x}^0, \mathbf{v}^0)$, there exist $\kappa_{low}(\mathbf{x}^0, \mathbf{v}^0) < \kappa_{upp}(\mathbf{x}^0, \mathbf{v}^0)$ such that

$$\begin{aligned} \kappa > \kappa_{upp} &\quad \Longrightarrow \quad \text{Flocking}, \\ \kappa \leq \kappa_{low} &\quad \Longrightarrow \quad \text{Non-flocking}. \end{aligned}$$

Problem: if there exists a κ_c such that

$$\begin{aligned} \kappa > \kappa_c \quad &\Longrightarrow \quad \text{Flocking}, \\ \kappa \le \kappa_c \quad &\Longrightarrow \quad \text{Non-flocking}. \end{aligned}$$

We call such κ_c the critical coupling strength.
It is open.

The local flocking

Definition: Multi-cluster flocking

We call an ensemble $\mathcal{G}$ tends to multi-cluster flocking aymptotically if there exist sub-classes $\mathcal{G}_\alpha = (\mathbf{x}_{\alpha i}, \mathbf{v}_{\alpha i}),\ \alpha = 1, \cdots, m$ such that

(i) $\mathcal{G} = \bigcup \mathcal{G}_\alpha, \quad |\mathcal{G}_\alpha| = N_\alpha \geq 1, \quad \sum_{\alpha=1}^m N_\alpha = N,$

(ii) $\sup_{t\geq 0} |\mathbf{x}_{\alpha i} - \mathbf{x}_{\alpha j}| < \infty, \quad \lim_{t\to 0} |\mathbf{v}_{\alpha i} - \mathbf{v}_{\alpha j}| = 0,$ for any $1 \leq \alpha \leq m$, and $1 \leq i, j \leq N_\alpha$,

(iii) $|\mathbf{x}_{\alpha i} - \mathbf{x}_{\beta j}| < \infty, \quad \liminf_{t\to 0} |\mathbf{v}_{\alpha i} - \mathbf{v}_{\alpha j}| > 0,$ for any $\alpha \neq \beta,\ 1 \leq i \leq N_\alpha$, and $1 \leq j \leq N_\beta$.

In particular, we call the ensemble $\mathcal{G}$ tends to bi-cluster flocking aymptotically when $m = 2$.

J. Cho, S.-Y. Ha, F. Huang, C. Jin, and D. Ko studied the bi-cluster flocking for the Cucker-Smale model with $\psi(s) = \frac{1}{(1+s)^\beta},\ \beta > 1$ [Cho, Ha, Huang, Jin, Ko, MMMAS, 2016]: Set

$$X(t) := \Big(\sum_{i=1}^{N_1} |\mathbf{x}_{1i}(t) - \mathbf{x}_{1c}(t)|^2\Big)^{\frac{1}{2}} + \Big(\sum_{i=1}^{N_2} |\mathbf{x}_{2i}(t) - \mathbf{x}_{2c}(t)|^2\Big)^{\frac{1}{2}},$$

$$V(t) := \Big(\sum_{i=1}^{N_1} |\mathbf{v}_{1i}(t) - \mathbf{v}_{1c}(t)|^2\Big)^{\frac{1}{2}} + \Big(\sum_{i=1}^{N_2} |\mathbf{v}_{2i}(t) - \mathbf{v}_{2c}(t)|^2\Big)^{\frac{1}{2}},$$

$$M_2(t) = \Big(\sum_{i=1}^{N} |\mathbf{v}_i(t)|^2\Big)^{\frac{1}{2}}, \quad \text{and} \quad \alpha_0(k) := \frac{1}{2}(\mathbf{v}_{1c}^k(0) - \mathbf{v}_{2c}^k(0)),$$

where $\mathbf{x}_{\alpha i}$, $\mathbf{v}_{\alpha i}$ is the position and velocity of the i-th particle in group $\mathcal{G}_\alpha$.

They showed, if

$$V_0 < \frac{\kappa \min\{N_1, N_2\}}{N} \int_{\mathcal{X}_0}^{\infty} \phi(\sqrt{2}s)ds, \quad \alpha_0(k) > 4\sqrt{2}V_0,$$

$$\min_{i,j}(\mathbf{x}_{1i}^k(0) - \mathbf{x}_{2j}^k(0)) > r_0(k), \quad \text{and} \quad \int_{r_0(k)}^{\infty} \phi(s)ds < \frac{\alpha_0(k)V_0}{2\sqrt{2}\kappa M_2(0)},$$

the bi-cluster flocking emerges asymptotically. More precisely, it holds that

$$\begin{aligned} &(i) \quad \min_{\alpha\neq\beta,i,j} |\mathbf{x}_{1i}(t) - \mathbf{x}_{2j}(t)| > r_0(k) + \frac{\alpha_0(k)t}{2}; \\ &(ii) \quad \exists X_M \text{ such that } X(t) \le X_M; \\ &(iii) \quad V(t) \le \mathcal{O}(1)(1+s)^{-\beta}. \end{aligned}$$

S.-Y. Ha, D. Ko, and Y. Zhang [Ha-Ko-Z, MMMAS, 2017] studied the multi-cluster flocking for the Cucker-Smale model with positive decreasing short-ranged communication weight: Set

$$X_\alpha(t) := \max_{i,\alpha} |\mathbf{x}_{\alpha i}(t) - \mathbf{x}_{\alpha c}(t)|, \quad X_\alpha(t) := \max_{i,\alpha} |\mathbf{v}_{\alpha i}(t) - \mathbf{v}_{\alpha c}(t)|,$$

$$\bar{\lambda}_0 := \frac{1}{2} \min_{\beta \neq \alpha} |\mathbf{v}^c_\beta(0) - \mathbf{v}^c_\alpha(0)|,$$

$$\text{and} \quad \int_{\bar{r}_0}^{\infty} \psi(s) ds := \frac{\gamma_N \bar{\lambda}_0}{8(1-\gamma_N)\sqrt{2} M_2(0)} \min_\alpha \int_{X_\alpha(0)}^{\infty} \psi(2s) ds,$$

where $\gamma_N := \frac{\min_\alpha N_\alpha}{N}$.

Then if

$$V_\alpha(0) < \frac{\bar{\lambda}_0}{4}, \quad \min_{\beta\neq\alpha,i,k}(\mathbf{x}^0_{\beta k} - \mathbf{x}^0_{\alpha i}) \cdot \frac{\mathbf{v}^c_\beta(0) - \mathbf{v}^c_\alpha(0)}{|\mathbf{v}^c_\beta(0) - \mathbf{v}^c_\alpha(0)|} > \bar{r}_0,$$

$$\text{and} \quad \kappa_1 < \kappa < \kappa_2,$$

where $\kappa_1 := \max\limits_\alpha \frac{2X_\alpha(0)}{\gamma_N \int_{X_\alpha(0)}^\infty \psi(2s)ds}$ and $\kappa_2 := \max\limits_\alpha \frac{2\bar{\lambda}_0}{3\gamma_N \int_{X_\alpha(0)}^\infty \psi(2s)ds}$. The multi-cluster flocking emerges asymptotically. More precisely, it holds that

$$(i) \quad \min_{\alpha\neq\beta,i,j} |\mathbf{x}_{\alpha i}(t) - \mathbf{x}_{\beta j}(t)| > \bar{r}_0 + \bar{\lambda}_0 t;$$

$$(ii) \quad \text{there exists } X^\infty_\alpha \text{ such that } X_\alpha(t) \leq X^\infty_\alpha;$$

$$(iii) \quad V_\alpha(t) \leq C \max\left\{ e^{-\frac{\kappa\phi(2X_\alpha(0))t}{2N}}, \phi\left(\bar{r}_0 + \frac{\bar{\lambda}_0 t}{2}\right)\right\},$$

where C is a positive constant.

Other results

Global flocking:

- the stochastic Cucker-Smale model:
  - F. Cucker, E. Mordecki, J. Math. Pures Appl.,2008;
  - S.-Y. Ha, K. Lee, and D. Levy, Commun. Math. Sci., 2009;
  - S. M. Ahn, and S.-Y. Ha, J. Math. Phys., 2010.
- the collision avoiding Cucker-Smale model:
  - F. Cucker, and J.-G. Dong, IEEE Trans. Automat. Control, 2010;
  - S. M. Ahn, H. Choi, S.-Y. Ha, and H. Lee, Commun. Math. Sci., 2012.
- $\cdots\cdots$
- Carrillo, Tadmor, Shvydkoy, $\cdots$

Local flocking:

- J. Cho, S.-Y, Ha, F. Huang, C. Jin, and D. Ko [Cho,etc, AA, 2016] studied the bi-cluster flocking in the Cucker-Smale model with unit speed;
- S.-Y. Ha, D. Ko, X. Zhang, and Y. Zhang [Ha-Ko-Z, JMP, 2017] studied the bi-cluster flocking in the kinetic Cucker-Smale model;
- L. Ru, and X. Xue [Ru-Xue, JFI, 2017] studied the number of cluster that the agents can be divided into if the initial position and velocity have ordering relations;
- S.-Y. Ha, D. Ko, and Y. Zhang [Ha-Ko-Z, DCDS, 2018] studied the multi-cluster flocking in the Cucker-Smale model with unit speed;
- Z. Qiao, Y. Liu, and X. Wang, [Qiao-Liu-Wang, JSSC, 2022] studied the multi-cluster flocking in the delayed Cucker-Smale model;
- J. Kim, and J.-H. Park, [Kim-Park,CSF, 2023] studied the multi-cluster flocking in the Cucker-Smale model with external force

A deficiency of Cucker-Smale model

If a small group of agents is located at a large distance from a large group of agents, the dynamics of the small group will be almost halted due to the small normalization pre-factor $\frac{\kappa}{N}$.

Motsch-Tadmor model

In order to remedy this deficiency, Motsch and Tadmor instead used $\Big(\sum_{k=1}^N \phi(|\mathbf{x}_k - \mathbf{x}_i|)\Big)^{-1}$ as a suitable normalization factor [Motsch, Tadmor, JSP, 2011]:

$$\text{(M-T Model)}\quad \begin{cases} \dot{\mathbf{x}}_i = \mathbf{v}_i, \\ \dot{\mathbf{v}}_i = \kappa \displaystyle\sum_{j=1}^N \psi_{ij}(\mathbf{v}_j - \mathbf{v}_i), \\ (\mathbf{x}_i, \mathbf{v}_i)\Big|_{t=0} = (\mathbf{x}_i^0, \mathbf{v}_i^0), \end{cases}$$

where

$$\psi_{ij} := \frac{\psi(|\mathbf{x}_j - \mathbf{x}_i|)}{\sum_{k=1}^N \psi(|\mathbf{x}_k - \mathbf{x}_i|)}.$$

However, the communication weight is not symmetric again:

$$\psi_{ij} \neq \psi_{ji},$$

this causes many mathematical difficulties, e.g., in the Cucker-Smale model, the symmetry implies the total momentum is conserved

$$\frac{d}{dt}\Big(\frac{1}{N}\sum_{i=1}^{N} v_i(t)\Big) = 0, \quad \Rightarrow \quad \overline{v}(t) := \frac{1}{N}\sum_{j=1}^{N} v_j = \overline{v}(0),$$

and the Cucker-Smale system is dissipative

$$\begin{aligned} \frac{d}{dt}\Big(\frac{1}{N}\sum_{i=1}^{N} |v_i - \overline{v}|^2\Big) &= -\frac{\kappa}{2N^2}\sum_{i,j} \psi_{ij}|v_i - v_j|^2 \\ &\leq -\min_{i,j}\phi_{ij} \times \frac{\kappa}{N}\sum_{i} |v_i - \overline{v}|^2. \end{aligned}$$

Consequently yields the large time behavior, $x \approx \overline{v}t$, and hence $\min_{ij} \phi_{ij}(t) \geq \phi(\overline{v}t)$, which gives

$$v_i(t) \to \overline{v}(0).$$

However, for the Motsch-Tadmor model, because of loss of symmetry, we loss such conservation of momentum and the dissipative property.

By introducing active sets, Motsch, Tadmorand showed, if

$$\kappa > \frac{d_V(0)}{\int_{d_X(0)}^{\infty} \psi^2(r)dr},$$

flocking emerge asymptotically. In particular, flocking emerge unconditionally if

$$\int_{d_X(0)}^{\infty} \psi^2(r)dr = \infty,$$

where

$$d_X(t) := \max_{1\le i,j\le N} |x_i(t) - x_j(t)|, \quad d_V(t) := \max_{1\le i,j\le N} |v_i(t) - v_j(t)|.$$

Liu and Xue studied the (M-T model) with $\psi(s) = (1+s)^{-2\beta}, \beta > 0$, and showed if [Li, Xue, DSA, 2015]

$$d_V(0) < \frac{\kappa}{N} \int_{d_X(0)}^{\infty} \phi(s) ds,$$

flocking emerge asymptotically, i.e.,

$$\sup_{0 \le t \le \infty} d_X(0) < \infty \quad \text{and} \quad \lim_{t \to \infty} d_V(0) = 0.$$

In particular, flocking emerge unconditionally if

$$\int_{d_X(0)}^{\infty} \psi(r) dr = \infty.$$

Recently, cooperate with S.-Y. Ha and C. Jin, we study the multi-cluster flocking for the (M-T model). More precisely, we consider the Cauchy problem for the following subsystems:

$$
\text{(M-T Model)'} \quad \begin{cases} \dot{\mathbf{x}}_{\alpha i} = \mathbf{v}_{\alpha i}, \quad t > 0, \quad \alpha \in [m], \quad i \in [N_\alpha], \\ \dot{\mathbf{v}}_{\alpha i} = \kappa \sum\limits_{j=1}^{N_\alpha} \phi_{ji}^{\alpha\alpha} (\mathbf{v}_{\alpha j} - \mathbf{v}_{\alpha i}) + \kappa \sum\limits_{\substack{\beta=1 \\ \beta \neq \alpha}}^{m} \sum\limits_{j=1}^{N_\beta} \phi_{ji}^{\beta\alpha} (\mathbf{v}_{\beta j} - \mathbf{v}_{\alpha i}), \\ (\mathbf{x}_{\alpha i}, \mathbf{v}_{\alpha i}) \Big|_{t=0} = (\mathbf{x}_{\alpha i}^0, \mathbf{v}_{\alpha i}^0), \end{cases}
$$

where $N = \sum\limits_{\alpha=1}^{m} N_\alpha$ is the total number of agents, and the communication weight $\phi_{ji}^{\gamma\alpha}$ is given by

$$
\phi_{ji}^{\gamma\alpha} := \frac{\phi(|\mathbf{x}_{\gamma j} - \mathbf{x}_{\alpha i}|)}{\sum\limits_{\beta=1}^{m} \sum\limits_{l=1}^{N_\beta} \phi(|\mathbf{x}_{\beta l} - \mathbf{x}_{\alpha i}|)}, \quad (\alpha, \gamma) \in [m] \times [m], \quad (i, j) \in [N_\alpha] \times [N_\gamma].
$$

It is easy to see that, for any $(\alpha, i) \in [m] \times [N_\alpha]$

$$\phi_{ji}^{\gamma\alpha} > 0 \quad \text{and} \quad \sum_{\gamma=1}^{m} \sum_{j=1}^{N_\gamma} \phi_{ji}^{\gamma\alpha} = \frac{\sum_{\gamma=1}^{m} \sum_{j=1}^{N_\gamma} \phi(|\mathbf{x}_{\gamma j} - \mathbf{x}_{\alpha i}|)}{\sum_{\beta=1}^{m} \sum_{l=1}^{N_\beta} \phi(|\mathbf{x}_{\beta l} - \mathbf{x}_{\alpha i}|)} = 1.$$

Then

$$\dot{\mathbf{v}}_{\alpha i} = -\kappa \mathbf{v}_{\alpha i} + \kappa \sum_{j=1}^{N_\alpha} \phi_{ji}^{\alpha\alpha} \mathbf{v}_{\alpha j} + \kappa \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^{m} \sum_{j=1}^{N_\beta} \phi_{ji}^{\beta\alpha} \mathbf{v}_{\beta j},$$

We use the property of $\phi_{ij}^{\gamma\alpha}$ to have

$$
\begin{aligned}
\frac{1}{\kappa}\dot{\mathbf{v}}_{\alpha b_t}(t) &= \sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha}\mathbf{v}_{\alpha j}(t) + \sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\mathbf{v}_{\beta j}(t) - \mathbf{v}_{\alpha b_t}(t)\\
&= \sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha}\left(\sum_{k=1}^{N_\alpha}\phi_{ks_t}^{\alpha\alpha} + \sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma}\phi_{ks_t}^{\gamma\alpha}\right)\mathbf{v}_{\alpha j}(t)\\
&\quad + \sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\left(\sum_{k=1}^{N_\alpha}\phi_{ks_t}^{\alpha\alpha} + \sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma}\phi_{ks_t}^{\gamma\alpha}\right)\mathbf{v}_{\beta j}(t) - \mathbf{v}_{\alpha b_t}(t),
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{1}{\kappa}\dot{\mathbf{v}}_{\alpha s_t}(t) &= \sum_{k=1}^{N_\alpha} \phi_{ks_t}^{\alpha\alpha}\mathbf{v}_{\alpha k}(t) + \sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma} \phi_{ks_t}^{\gamma\alpha}\mathbf{v}_{\gamma k}(t) - \mathbf{v}_{\alpha s_t}(t) \\
&= \sum_{k=1}^{N_\alpha} \phi_{ks_t}^{\alpha\alpha}\left(\sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha} + \sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\right)\mathbf{v}_{\alpha k}(t) \\
&\quad + \sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma} \phi_{ks_t}^{\gamma\alpha}\left(\sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha} + \sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\right)\mathbf{v}_{\gamma k}(t) - \mathbf{v}_{\alpha s_t}(t).
\end{aligned}
$$

Hence, we can obtain

$$
\begin{aligned}
&\frac{1}{\kappa}(\dot{\mathbf{v}}_{\alpha b_t}(t)-\dot{\mathbf{v}}_{\alpha s_t}(t))\\
&\quad=-(\mathbf{v}_{\alpha b_t}(t)-\mathbf{v}_{\alpha s_t}(t))+\sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha}\sum_{k=1}^{N_\alpha}\phi_{ks_t}^{\alpha\alpha}(\mathbf{v}_{\alpha j}(t)-\mathbf{v}_{\alpha k}(t))\\
&\quad+\sum_{j=1}^{N_\alpha}\phi_{jb_t}^{\alpha\alpha}\sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma}\phi_{ks_t}^{\gamma\alpha}(\mathbf{v}_{\alpha j}(t)-\mathbf{v}_{\gamma k}(t))\\
&\quad+\sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\sum_{k=1}^{N_\alpha}\phi_{ks_t}^{\alpha\alpha}(\mathbf{v}_{\beta j}(t)-\mathbf{v}_{\alpha k}(t))\\
&\quad+\sum_{\substack{\beta=1\\ \beta\neq\alpha}}^{m}\sum_{j=1}^{N_\beta}\phi_{jb_t}^{\beta\alpha}\sum_{\substack{\gamma=1\\ \gamma\neq\alpha}}^{m}\sum_{k=1}^{N_\gamma}\phi_{ks_t}^{\gamma\alpha}(\mathbf{v}_{\beta j}(t)-\mathbf{v}_{\gamma k}(t)).
\end{aligned}
$$

Set

$$\begin{cases} \mathcal{D}(X_\alpha) := \max\limits_{1\le i\ne j\le N_\alpha} |\mathbf{x}_{\alpha i} - \mathbf{x}_{\alpha j}|, \\ \mathcal{D}(V_\alpha) := \max\limits_{1\le i\ne j\le N_\alpha} |\mathbf{v}_{\alpha i} - \mathbf{v}_{\alpha j}|, \\ \mathfrak{N}(X) := \min\limits_{\alpha\ne\beta, i, j} |\mathbf{x}_{\alpha i} - \mathbf{x}_{\beta j}|. \end{cases}$$

Note that

$$\begin{aligned} \sum_{k=1}^{N_\gamma} \phi_{ki}^{\gamma\alpha} &= \frac{\sum_{k=1}^{N_\gamma} \phi(|\mathbf{x}_{\gamma k} - \mathbf{x}_{\alpha i}|)}{\sum_{\gamma=1}^{m} \sum_{k=1}^{N_\gamma} \phi(|\mathbf{x}_{\gamma k} - \mathbf{x}_{\alpha i}|)} \\ &\le \frac{\sum_{k=1}^{N_\gamma} \phi(|\mathbf{x}_{\gamma k} - \mathbf{x}_{\alpha i}|)}{\sum_{k=1}^{N_\alpha} \phi(|\mathbf{x}_{\alpha k} - \mathbf{x}_{\alpha i}|)} \le \frac{N_\gamma \phi(\mathfrak{N}(X))}{N_\alpha \phi(\mathcal{D}(X_\alpha))}. \end{aligned}$$

we can deduce

$$\begin{cases} \dfrac{d}{dt}\mathcal{D}(X_\alpha) \le \mathcal{D}(V_\alpha), \quad \text{a.e., } t > 0, \\ \dfrac{d}{dt}\mathcal{D}(V_\alpha) \le -\dfrac{\kappa\phi(\mathcal{D}(X_\alpha))}{N}\mathcal{D}(V_\alpha) + 3\kappa\mu\mathcal{D}(V^0)\dfrac{\phi(\mathfrak{N}(X))}{\phi(\mathcal{D}(X_\alpha))}. \end{cases}$$

To get the decay of $\mathcal{D}(V_\alpha)$, we need to control $\mathcal{D}(X_\alpha)$ and hope $\phi(\mathfrak{N}(X))$ tends to infinity. For this, we introduce

$$\begin{cases} \mathfrak{M}(X, V^0) := \min\limits_{\alpha \neq \beta, i, j} \left\{ (\mathbf{x}_{\alpha i} - \mathbf{x}_{\beta j}) \cdot \mathbf{e}^0_{\alpha\beta} \right\}, \\ \mathfrak{M}(V, V^0) := \min\limits_{\alpha \neq \beta, i, j} \left\{ (\mathbf{v}_{\alpha i} - \mathbf{v}_{\beta j}) \cdot \mathbf{e}^0_{\alpha\beta} \right\}. \end{cases}$$

And a priori assume

$$\mathcal{D}(X_\alpha(t)) < 3 \max_\alpha \mathcal{D}(X^0_\alpha),$$

$$\text{and} \quad \mathfrak{M}(V(t), V^0) > \frac{\mathfrak{M}(V^0, V^0)}{3} > 0,$$

for $t \in [0, \tau)$. Then, we can show the emergence of flocking in $[0, \tau)$.

Hence, the remaining thing is to show $\tau = \infty$. For this, we introduce a class of admissible initial data

- $(\mathcal{F}1)$: (Sufficient concentration around cluster local averages)

$$\max_{\alpha} \mathcal{D}(V_{\alpha}^{0}) < \frac{\kappa}{N} \phi\Big(3 \max_{\alpha} \mathcal{D}(X_{\alpha}^{0})\Big) \max_{\alpha} \mathcal{D}(X_{\alpha}^{0}).$$

- $(\mathcal{F}2)$: (Sufficiently close to a well-separated multi-cluster configuration)

$$\mathfrak{M}(X^0, V^0) \geq 0,$$

$$\mathfrak{M}(V^0, V^0) > \max \left\{ 6\kappa \max_{\alpha} \mathcal{D}(X_{\alpha}^{0}),\ \frac{72 N \mu \mathcal{D}(V^0) \int_{\mathfrak{M}(X^0, V^0)}^{\infty} \phi(s) ds}{\Big(\max_{\alpha} \mathcal{D}(X_{\alpha}^{0})\Big) \phi^2 \Big(3 \max_{\alpha} \mathcal{D}(X_{\alpha}^{0})\Big)} \right\}$$

where $\mu = \max_{\alpha} \left(\frac{N}{N_{\alpha}} - 1\right)$.

and we show

Theorem[Ha-Jin-Z, submitted]

Suppose the framework $(\mathcal{F}1) - (\mathcal{F}2)$ holds, and let $\{(\mathbf{x}_{\alpha i}, \mathbf{v}_{\alpha i})\}$ be a global solution of (M-T model)'. Then multi-cluster flocking emerges asymptotically: for all $t \geq 0$ and $\alpha \in [m]$,

$$(i)\ \sup_{t\geq 0} \mathcal{D}(X_\alpha(t)) \leq 3 \max_\alpha \mathcal{D}(X_\alpha^0),$$

$$(ii)\ \mathcal{D}(V_\alpha(t)) \leq \mathcal{O}(1) \max \left\{ e^{-\frac{\kappa\phi(3 \max_\alpha \mathcal{D}(X_\alpha^0))t}{2N}}, \phi\left(\mathfrak{M}(X^0, V^0) + \frac{\mathfrak{M}(V^0, V^0)t}{6} \right) \right.$$

$$(iii)\ \mathfrak{N}(X(t)) > \mathfrak{M}(X^0, V^0) + \frac{\mathfrak{M}(V^0, V^0)t}{3}.$$

Thanks for your attention!